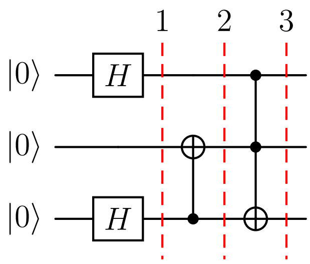
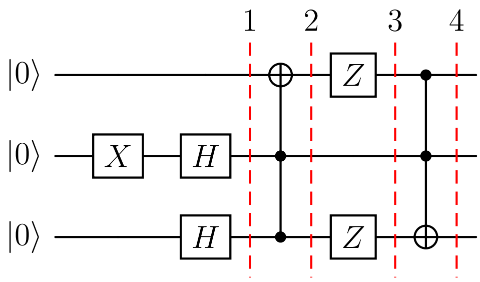
Sample of the 2nd IQC test

(This is just a trial test.)

1. Write down the state of the quantum circuits in the marked lines.

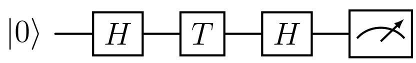


2. Write down the state of the quantum circuits in the marked lines.



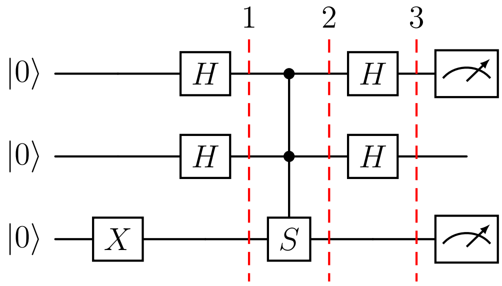
3. Create an orthonormal basis set to the computational basis $|0\rangle, |1\rangle$. Orthonormal means, that the basis vectors are normalized and orthogonal to each other (*jsou normované a navzájem kolmé*). How can you check the upper define orthogonality? How can we change from computational basis to your basis?

4. What is the probability of measuring 0 as the outcome of the following circuit?



5. Explain the uniform superposition (*rovnoměrná superpozice*). What is it good for?

6. Write down the state of the quantum circuits in the marked lines. Whats the probabilities of measuring the $|0\rangle$ and $|1\rangle$ in the qubits marked with the measurement symbol?



Solution with notes

General notes

The second exam will be focused on the quantum circuit analysis step by step. I strongly recommend to went through all materials from the exercises.

Approach to solve multiple qubit systems

The effect of multiple one-qubit gates can be written in the following way. (The first three gates were considered together, like when the X gate would be moved under H gates. It is just the formating.)

$$H \otimes H \otimes X |000\rangle = H |0\rangle \otimes H |0\rangle \otimes X |0\rangle$$

The effect of quantum gates is linear:

$$H \otimes H \otimes X \frac{1}{\sqrt{2}} \big(|000\rangle + |111\rangle \big) = \frac{1}{\sqrt{2}} \Bigg(H |0\rangle \otimes H |0\rangle \otimes X |0\rangle + H |1\rangle \otimes H |1\rangle \otimes X |1\rangle \Bigg)$$

Solution to task number 1:

$$|\psi_1\rangle = |+\rangle \otimes |0\rangle \otimes |+\rangle = \frac{1}{2} (|000\rangle + |001\rangle + |100\rangle + |101\rangle)$$

$$|\psi_2\rangle = \frac{1}{2} (|000\rangle + |011\rangle + |100\rangle + |111\rangle)$$

$$|\psi_3\rangle = \frac{1}{2} (|000\rangle + |011\rangle + |100\rangle + |110\rangle)$$

Solution to task number 2:

$$|\psi_1\rangle = |0\rangle \otimes |-\rangle \otimes |+\rangle = \frac{1}{2} (|000\rangle + |001\rangle - |010\rangle - |011\rangle)$$

$$|\psi_2\rangle = \frac{1}{2} (|000\rangle + |001\rangle - |010\rangle - |111\rangle)$$

Recall the effect of Z-gate on general one-qubit states.

$$|\psi_3\rangle = \frac{1}{2} (|000\rangle - |001\rangle - |010\rangle - |111\rangle)$$

$$|\psi_4\rangle = \frac{1}{2} (|000\rangle - |001\rangle - |010\rangle - |110\rangle)$$

Solution to task number 3:

A new orthonormal basis can be created by applying a suitable unitary transformation on all "old" basis states. As an example, Hadamard gate maps computational basis set (Z basis), into X basis$|0\rangle, |1\rangle \to |+\rangle, |-\rangle$. Quantum Fourier transform is another example applicable mapping the computational basis set to Fourier basis set. By computing the dot product (*skalární součin* one can check the orthogonality and also the norm of basis vectors, which have to be 1.

Solution to task number 4:

This is the interferometer with the specific value of φ, feel free to look in the materials from the exercises.

Solution to task number 5:

I will let this answer for self-study :)

Solution to task number 6:

This task was too complicated and long to be in the exam. Similar, yet easier and smaller circuit might appear in exam. Especially the calculation of probability measuring one qubit out of several in the circuit.

Slice 1

$$|\psi_1\rangle = H \otimes H \otimes X\,|000\rangle = H\,|0\rangle \otimes H\,|0\rangle \otimes X\,|0\rangle$$

$$|\psi_1\rangle = |+\rangle \otimes |+\rangle \otimes |1\rangle$$

$$\boxed{|\psi_1\rangle = \frac{1}{2}\big(\,|001\rangle + |011\rangle + |101\rangle + |111\rangle\,\big)}$$

Slice 2

The gates here form a controlled-controlled-S:

Control on $q_0 = 1$, control on $q_1 = 1$, S acting on the bottom qubit:

$$S\,|1\rangle = i\,|1\rangle\,.$$

Only the basis state $|111\rangle$ satisfies both controls, so:

$$|111\rangle \longrightarrow i\,|111\rangle\,.$$

Thus:

$$\boxed{|\psi_2\rangle = \frac{1}{2}\big(\,|001\rangle + |011\rangle + |101\rangle + i\,|111\rangle\,\big)}.$$

Slice 3: Apply $H \otimes H$ on the First Two Qubits

We apply $H \otimes H \otimes I$ to each term of $|\psi_2\rangle$. The detailed expansion consists of 16 elements and I don't want to specify it here. The probability amplitudes (coefficients of elements) can be added, some subtracted, the final results is:

$$\boxed{|\psi_3\rangle = \frac{1}{4}\Big[(3+i)\,|001\rangle + (1-i)\,|011\rangle + (1-i)\,|101\rangle + (-1+i)\,|111\rangle\Big]}.$$

Only these four basis states have nonzero amplitude.

Measurement Probabilities

We measure top and bottom qubit.

Bottom qubit

All four basis states $|001\rangle, |011\rangle, |101\rangle, |111\rangle$ end with 1:

$$P(q_2 = 1) = 1, \qquad P(q_2 = 0) = 0.$$

Top qubit

Group probabilities:

$$P(|001\rangle) = \frac{|3+i|^2}{16} = \frac{10}{16} = \frac{5}{8},$$
$$P(|011\rangle) = \frac{|1-i|^2}{16} = \frac{2}{16} = \frac{1}{8},$$
$$P(|101\rangle) = \frac{|1-i|^2}{16} = \frac{1}{8},$$
$$P(|111\rangle) = \frac{|-1+i|^2}{16} = \frac{2}{16} = \frac{1}{8}.$$

States with $q_0 = 0$:

$$P(q_0 = 0) = \frac{5}{8} + \frac{1}{8} = \frac{3}{4}.$$

States with $q_0 = 1$:

$$P(q_0 = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

$$\boxed{P(q_0 = 0) = \frac{3}{4}, \qquad P(q_0 = 1) = \frac{1}{4}.}$$